Fiscal Year:	FY 2009	**Task Last Updated:**	FY 03/23/2012
PI Name:	Albert, Mitchell S Ph.D.		
Project Title:	Hyperpolarized 129Xe MRI of the Brain, Lungs and Plants Augmented with MRI Pulse Sequence Development		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Physiology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	01655	**Congressional District:**	3
Comments:	Note: PI moved to U Mass Medical School in 2007; previously at Brigham and Women's Hospital/Harvard University (9/2008)		
Project Type:	Ground	**Solicitation / Funding Source:**	PECASE AWARD
Start Date:	09/18/2007	**End Date:**	09/17/2009
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received no-cost extension to 9/17/2009 (from 9/17/2008) per J. Dardano/JSC (9/2008)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Reilly, John Joseph (Brigham and Women's Hospital) Mulkern , Robert (Children's Hospital Boston)		
Grant/Contract No.:	NNX07AP80G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We are developing a new diagnostic technology: magnetic resonance imaging (MRI) and spectroscopy of hyperpolarized noble gases. Large non-equilibrium nuclear spin polarizations can be created in the spin-1/2 noble gases, 3He and 129Xe, using the technique of spin-exchange optical pumping. Such large polarizations greatly enhance the magnetic resonance detection of 3He and 129Xe, enabling high-resolution MRI studies of the lung and airways, as well as dissolved tissue-phase investigations of organs and tissues, such as the brain, using lipophilic 129Xe. An important feature of hyperpolarized noble gas MR is that it does not require a large magnetic field, unlike the conventional water proton MR method used in medical MRI. Low magnetic field MRI with hyperpolarized noble gases can allow us to use a small, light-weight, low-power devices. This capability provides potential for advances in both portable ground-based and practical space-based MR systems. Our proposed ground-based investigations with hyperpolarized noble gas MR will affect biomedical research in two important ways: (1) by providing the potential to acquire novel information about physiological structure and function of the lung and the brain in space-flight and in terrestrial gravity, and (2) by developing a new diagnostic technology that has clinical applications for diagnosis of pathology.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	An important feature of hyperpolarized noble gas MR is that it does not require a large magnetic field, unlike the conventional thermally polarized water proton MR method used in medical MRI. As a result, at low magnetic fields (0.01 Tesla) the signal-to-noise ratio (SNR) of hyperpolarized noble gas MR is much larger than the SNR of conventional thermally polarized MR. Low field MR utilizing hyperpolarized noble gases should be practical in a small, light-weight, low-power device, providing the potential for both portable ground-based and practical space-based MR systems. Specific magnetic and biological properties of hyperpolarized 3He and 129Xe MRI include: (i) airway and lung gas space imaging (for diagnosis of chronic obstructive pulmonary disease (COPD), e.g. emphysema, asthma, and for studies of respiratory function); (ii) the imaging of lipid structures in the brain (for the diagnosis and staging of demyelinating diseases such as multiple sclerosis, and in research on brain function); and (iii) angiography and measurement of blood flow (for diagnosis of stroke and ischemia, and for research into the function of the brain and cardiopulmonary system). The recent progress made in our lab lays the foundation for the continued development of hyperpolarized xenon MRI for brain imaging. Specifically, imaging of lipid structures in the brain (for the diagnosis and staging of demyelinating diseases such as multiple sclerosis) and angiography and measurement of blood flow (for diagnosis of stroke and ischemia, and for research into the function of the brain and cardiopulmonary system), are now viable research goals. Because HP 129Xe is an MR signal source with properties very different from those generated from water-protons, it may yield structural and functional information not detectable by proton-based MRI methods. Thus brain imaging with hyperpolarized xenon should be of great benefit to the diagnosis and treatment monitoring of a number of neurological diseases.
Task Progress:	[Ed. note 3/23/2012: compiled from Final Report submitted to JSC Human Research Program in July 2011 and received by Task Book February 2012] In work by the PI and others, HP noble gas MRI using 3He and 129Xe has been established to provide highly detailed images of the lungs and brain, and offers the immense advantage of providing information on physiological function, including lung ventilation (3He or 129Xe), lung perfusion (129Xe), and cerebral blood flow (129Xe). For example, HP gas MRI allows investigators to pinpoint precise sections of the lungs where gas is reaching, and where gas is not reaching (ventilation defects), something other measurements of lung function cannot do. Because of its greater sensitivity, HP gas MRI can detect pulmonary or cerebral dysfunction at earlier stages, allowing treatment or counteracting measures to be provided earlier and with greater effectiveness. In addition, HP gas MRI provides imaging of physiological function without exposing subjects to ionizing radiation or chemical contrast agents. This increases safety, and also allows investigators and/or care providers to non-invasively image subjects repeatedly over time. In this study, our specific research accomplishments included the following: Extending the Frequency of the 1.5 T Signa MRI Scanner to 3He Human Lung Imaging Using HP 3He Development of Imaging Pulse Sequences for HP Gas MRI Small Airway Imaging of the Human Lungs Human 3D Airway Tree Rendering Dynamically Adaptive MRI Optical-Flow Imaging of Rat Lung Dynamics HP 129Xe Lifetimes (T1) in Blood Temporal Dynamics of 129Xe in Living Rats Cerebral Perfusion Measurement Using HP 129Xe
Bibliography Type:	Description: (Last Updated: 03/23/2012)
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